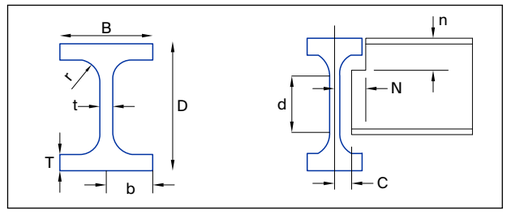
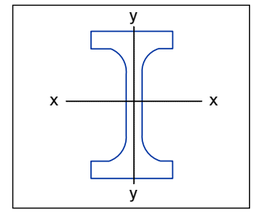
2.3
Universal Beams & Columns (UB & UC)



Imperial Units

Designation			Depth of Section	Width of Section	Thickness		Root Radius	Depth between Fillets	Area of Section	Ratios for Local Buckling		Dimensions for Detailing		
Size	Mass per Metre				Flange	Web				Flange	Web	End clearance	Notch	
			D	B	T	t	r	d	A	b/T	d/t	C	N	n
in (mm)	lb/ft	kg/m	mm	mm	mm	mm	mm	mm	cm²			mm	mm	mm
W18-18x6 (457x152)	46	68.5	458.7	153.9	15.3	9.1	10.2	407.7	87.1	5.03	44.6	7	82	26
	50	74.2	462.0	154.4	17.0	9.6	10.2	407.6	94.5	4.54	42.5	7	82	27
	55	82.1	465.8	155.3	18.9	10.5	10.2	407.6	105	4.11	38.8	7	82	29
W18-18x7 1/2 (457x191)	45	67.1	453.4	189.9	12.7	8.5	10.2	407.6	85.5	7.48	48.0	6	101	23
	50	74.3	457.0	190.4	14.5	9.0	10.2	407.6	94.6	6.57	45.3	7	101	25
	55	82.0	460.0	191.3	16.0	9.9	10.2	407.6	104	5.98	41.2	7	101	26
	60	89.3	463.4	191.9	17.7	10.5	10.2	407.6	114	5.42	38.8	7	101	28
	65	96.7	466.1	192.8	19.1	11.4	10.2	407.6	123	5.06	35.7	8	101	29
	66	98.3	467.2	192.8	19.6	11.4	10.2	407.6	125	4.92	35.8	8	101	30
	71	105.7	469.1	193.9	20.6	12.6	10.2	407.6	134	4.71	32.4	8	101	31
W18-18x11 (457x279)	76	113.1	462.5	280.3	17.3	10.8	10.2	407.6	144	8.12	37.7	7	145	27
	86	128	467.1	281.7	19.6	12.1	10.2	407.6	163	7.20	33.7	8	145	30
	97	144.4	472.2	283.1	22.1	13.6	10.2	407.6	184	6.40	30.0	9	145	32
	106	157.7	475.7	284.5	23.9	15.0	10.2	407.5	201	5.96	27.2	9	145	34
	119	177.1	481.8	286.1	26.9	16.6	10.2	407.6	226	5.31	24.5	10	145	37
	130	193.5	489.0	283.5	30.5	17.0	10.2	407.6	247	4.65	24.0	11	143	41
	143	212.8	495.0	285.0	33.5	18.5	10.2	407.5	271	4.25	22.0	11	143	44
W21-21x6 1/2 (533x165)	44	65.5	524.8	165.1	11.4	8.9	12.7	476.6	84	7.24	53.6	6	88	24
	50	74.4	529.1	165.9	13.5	9.7	12.7	476.7	95	6.14	49.4	7	88	26
	57	84.8	534.9	166.5	16.5	10.3	12.7	476.5	108	5.04	46.3	7	88	29
W21-21x8 1/4 (533x210)	55	82.2	528.3	208.8	13.2	9.6	12.7	476.5	105	7.91	49.6	7	110	26
	62	92.1	533.1	209.3	15.6	10.1	12.7	476.5	117	6.71	47.2	7	110	28
	68	101	536.7	210.0	17.4	10.8	12.7	476.5	129	6.03	44.1	7	110	30
	73	109	539.5	210.8	18.8	11.6	12.7	476.5	139	5.61	41.1	8	110	32
	82	122	544.5	211.9	21.3	12.7	12.7	476.5	155	4.97	37.5	8	110	34
	83	123.5	544.3	212.2	21.2	13.1	12.7	476.5	157	5.00	36.4	9	110	34
	93	138.4	549.1	213.9	23.6	14.7	12.7	476.5	176	4.53	32.3	9	110	36
W21-21x12 1/4 (533x312)	101	150.3	542.5	312.2	20.3	12.7	12.7	476.5	192	7.68	37.5	8	160	33
	111	165.2	546.4	313.4	22.2	14.0	12.7	476.6	211	7.05	34.1	9	160	35
	122	181.6	550.7	314.7	24.4	15.2	12.7	476.5	231	6.45	31.3	10	160	37
	132	196.4	554.5	316.0	26.3	16.5	12.7	476.5	250	6.01	28.9	10	160	39
	147	218.8	560.3	317.8	29.2	18.3	12.7	476.5	279	5.44	26.1	11	160	42
	166	247	571.0	315.5	34.5	19.1	12.7	476.5	315	4.57	25.0	12	158	47
	182	270.8	577.1	317.8	37.6	21.1	12.7	476.5	346	4.23	22.6	13	158	50

Dimensions	EN10365 / ASTM A6
Specification	EN10025 / ASTM A36 / ASTM A572
Size Range	UB: 5”x3” (127mm x 76mm) to 44”x16” (1118mm x 406mm) UC: 4”x4” (102mm x 102mm) to 14”x16” (356mm x 406mm)



Surface Area	Second Moment of Area		Radius of Gyration		Elastic Modulus		Plastic Modulus		Buckling Parameter	Torsional Index	Warping Constant	Torsional Constant
Per metre	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y				
									u	x	H	J
m²	cm⁴	cm⁴	cm	cm	cm³	cm³	cm³	cm³			dm⁶	cm⁴
1.50	29530	933	18.4	3.27	1288	121	1482	191	0.869	33.0	0.458	50.4
1.50	32670	1047	18.6	3.33	1414	136	1627	213	0.873	30.1	0.518	65.9
1.51	36590	1185	18.7	3.37	1571	153	1811	240	0.873	27.4	0.591	89.2
1.63	29380	1452	18.5	4.12	1296	153	1471	237	0.872	37.9	0.705	37.1
1.64	33320	1671	18.8	4.20	1458	176	1653	272	0.877	33.9	0.818	51.8
1.65	37050	1871	18.8	4.23	1611	196	1831	304	0.877	30.9	0.922	69.2
1.66	41020	2089	19.0	4.29	1770	218	2014	338	0.880	28.3	1.04	90.7
1.66	44590	2281	19.0	4.30	1913	237	2184	369	0.879	26.3	1.14	114
1.67	45730	2347	19.1	4.33	1957	243	2232	379	0.881	25.7	1.18	121
1.67	48760	2507	19.0	4.32	2079	259	2383	404	0.877	24.4	1.26	145
2.01	55460	6344	19.6	6.64	2398	453	2669	692	0.885	27.9	3.14	118
2.02	63520	7295	19.7	6.69	2720	518	3039	792	0.886	24.8	3.65	170
2.03	72710	8367	19.9	6.74	3079	591	3457	906	0.886	22.1	4.24	244
2.04	79600	9178	19.9	6.76	3347	645	3775	991	0.884	20.5	4.68	311
2.06	91040	10530	20.1	6.82	3779	736	4284	1132	0.885	18.4	5.44	442
2.06	102500	11590	20.4	6.86	4192	818	4761	1257	0.891	16.6	6.09	603
2.08	114400	12960	20.5	6.91	4624	910	5278	1400	0.891	15.2	6.90	800
1.67	35030	859	20.5	3.20	1335	104	1561	166	0.847	47.0	0.566	31.9
1.68	40810	1032	20.8	3.30	1543	124	1797	199	0.853	41.4	0.686	47.1
1.69	48640	1276	21.2	3.44	1819	153	2107	243	0.862	35.5	0.857	73.8
1.85	47540	2008	21.3	4.38	1800	192	2059	300	0.864	41.6	1.33	51.5
1.86	55230	2389	21.7	4.51	2072	228	2360	356	0.872	36.5	1.60	75.7
1.87	61520	2692	21.9	4.57	2292	256	2612	399	0.874	33.2	1.82	101
1.88	66820	2943	21.9	4.60	2477	279	2828	436	0.875	30.9	1.99	126
1.89	76040	3388	22.1	4.67	2793	320	3196	500	0.877	27.6	2.32	178
1.89	76240	3389	22.0	4.65	2801	319	3212	500	0.874	27.6	2.32	181
1.90	86170	3868	22.1	4.68	3139	362	3617	569	0.873	24.9	2.67	251
2.29	100800	10320	22.9	7.33	3715	661	4147	1012	0.885	27.8	7.03	217
2.30	111300	11410	23.0	7.36	4074	728	4565	1117	0.885	25.6	7.84	284
2.31	123300	12680	23.1	7.40	4476	806	5032	1238	0.885	23.4	8.78	373
2.32	134200	13850	23.2	7.44	4842	876	5462	1348	0.884	21.8	9.66	469
2.33	151200	15650	23.3	7.49	5396	985	6116	1519	0.884	19.8	11.0	643
2.34	178000	18110	23.8	7.58	6233	1148	7080	1766	0.891	17.2	13.0	983
2.36	197200	20150	23.9	7.63	6834	1268	7807	1956	0.890	15.9	14.7	1278